



Equity Preservation Mortgage

Technical Architecture — Version 5.0

FutureProof Financial Group Limited

Income from Property Equity

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1. EPM Overview

FutureProof transforms home equity into monthly income. We partner with homeowners, borrowers, and capital providers to build a simple system: invest equity, distribute returns, manage risk.

The Four Players

Role	What They Do	Their Stake
Customer	Partners their equity for monthly income	Receives monthly distributions
Investment Partner	Manages the invested capital	Generates ~10% p.a. returns over 30 years
Lender	Originates loans, manages customer relationships	Earns margin (0.75%–2%) on distributions
Wholesale Funder	Supplies capital to lenders	Earns wholesale margin (0.2%–3.6%)

How Monthly Income Works

1. Investment Partner's fund earns returns.
2. System calculates monthly payout (based on equity %, fund performance, holiday status).
3. Customer gets their share.
4. Lender deducts margin. Wholesale Funder takes their cut. FutureProof takes 0.25%.

Simple. Repeatable. Auditable.

2. Customer Journey

Phase 1: Origination (Weeks 1–4)

1. Customer submits application.
2. Deal Agent verifies identity (KYC/AML).
3. Property valued and risk tier assigned.
4. Customer e-signs documents.
5. Lender and funder approve the deal.
6. Capital transferred.

7. Account goes live.

Phase 2: Ongoing (Months 1–30+)

Every month:

- 1. Distribution Agent calculates fund returns.
- 2. Customer gets paid on fixed date.
- 3. Investment Agent monitors fund performance — if it drops 10%, interest holiday triggers automatically.
- 4. Monthly statement issued showing distribution, margins, fund health, and holiday status.
- 5. Customer can request early exit at any time.

Phase 3: Exit (Early or Maturity)

- 1. Customer requests exit.
- 2. Deal Agent calculates final amount (fund value minus fees).
- 3. Lender approves.
- 4. Distribution Agent processes final payment.
- 5. Loan closed.
- 6. All records archived (7 years).

3. Stakeholder Views & Fraud Monitoring

Stakeholder	Dashboard Shows	Red Flags
Customer	Monthly distributions, cumulative income, fund chart, holiday status	Multiple withdrawals same day, logins from new locations, rapid device switches
Lender	Distribution schedule, margin received, portfolio health, customer status	Holiday requests that look systematic, unusual pattern shifts, distributions delayed
Investment Partner	Fund performance, ETF data, rebalancing schedule, cost basis	Performance that doesn't match ETFs, holdings mismatches, rebalancing delays
Wholesale Funder	Portfolio yield, default rates, margin collections, lender tier performance	Yield compression, risk concentration, lender financial strain

4. System Architecture

Core Platform

Rails 8.1 + Hotwire + Stimulus

Single monolithic Rails app. Live dashboards update via Hotwire (Turbo). Lightweight JavaScript via Stimulus. No WebSocket complexity.

PostgreSQL

- Source of truth for customers, transactions, and audit trails.
- JSON columns for flexible fund metadata.
- Materialized views for performance aggregation.
- Database triggers enforce compliance rules automatically.
- Partitioned by customer ID for scale.

Kafka Event Queue

Agents publish events; others subscribe. Exactly-once processing with dead-letter queues for failures.

1. Deal created → Distribution Agent and Compliance Agent react.
2. Distribution executed → Lender gets paid.
3. Holiday triggered → Distribution Agent adjusts calculations.

Redis Cache

Sessions, rate limiting, staged calculations, fraud detection (Bloom filters), frequently accessed data.

S3 Storage

- Documents, statements, audit logs.
- Encrypted at rest.
- Lifecycle: archive after 90 days, delete after 7 years (compliance).

Core Services

Distribution Engine (*Solid Queue job*)

1. Runs 1st–5th of each month via Rails native background job scheduler.
2. Pulls fund returns from Investment Partner.
3. Calculates per-customer distributions.
4. Stages results in Redis for validation.
5. Executes ACH/bank transfers via Stripe.
6. Publishes completion events to Kafka.

Retry logic with exponential backoff on failure.

Compliance Engine

- Real-time validation against 20+ regulatory rules.
- Writes immutable audit trail.
- Database triggers for Change Data Capture.

Fraud Detection

ML pipeline scores every transaction across three dimensions:

- **Known patterns** — Bloom filter matching.
- **Behavioral anomalies** — Isolation forest model.
- **Rule violations** — Impossible geography, velocity checks.

Interest Holiday Logic

- Monthly job monitors fund performance vs. entry/exit thresholds.
- Triggers automatically at 90% drop.
- Releases automatically at 145.8% recovery.

Notification Service

- Email (SendGrid), SMS (Twilio), in-app.
- Stores pending notifications, retries with backoff, confirms delivery.

Document Engine

- Server-rendered templates (ERB) + Prawn gem.
- Generates loan docs, agreements, statements, disclosures.
- All digitally signed, archived to S3 with immutable version IDs.

5. Four-Agent Architecture

Agent	Responsibilities	Triggered By
Deal Agent	Processes applications, runs KYC/AML, generates documents	New application
Distribution Agent	Calculates monthly returns, deducts margins, executes payments, generates statements	Monthly cutoff date
Investment Agent	Monitors fund performance, calculates rolling metrics, decides holiday eligibility, triggers/releases holidays	Monthly performance updates
Compliance Agent	Real-time fraud detection, regulatory rule enforcement, generates audit reports, flags suspicious activity	Transaction submission

Agent Communication

All agents publish to Kafka. Others subscribe. Decoupled. Asynchronous. Scalable.

Example Flow

1. Deal Agent publishes `deal.created` .
2. Distribution Agent and Compliance Agent subscribe and wake up.
3. Distribution Agent publishes `distribution.monthly` .
4. Lender receives payment.
5. Investment Agent publishes `holiday.triggered` .
6. Distribution Agent adjusts calculations.
7. Compliance Agent publishes `fraud.detected` .
8. System escalates to lender for review.

6. Agent Technology: Claude & Kimi

Why Agents?

Human agents would slow everything down. AI agents process in seconds. Four autonomous agents handle all workflows 24/7.

How We Do It: Claude & Kimi

Claude handles Deal Agent workflows:

- KYC/AML logic and eligibility assessment.
- Document generation decisions.

Kimi handles Distribution Agent workflows:

- Complex calculation logic.
- Margin deduction rules.
- Payment reconciliation.

Both are integrated via OpenAI-compatible API endpoints inside our private network.

Critical: No customer data ever leaves the system. All processing happens on-premise. Claude and Kimi receive anonymised, hashed transaction patterns and rule-based queries. They respond with recommendations that agents execute, but they never touch raw customer data.

Implementation Details

- Rails service classes (e.g., `DealAgent::OnboardingService`).
- Each agent runs as Solid Queue background job (4 workers per pod).
- Horizontally scalable.

Event Subscription (Kafka)

- Each agent subscribes to relevant Kafka topics using ruby-kafka gem.
- Consumer groups handle partitioning automatically.
- Dead-letter queues capture failures.

Idempotency & Retry

1. Agents implement idempotent handlers.
2. Database unique constraints + transactional locks prevent double-processing.
3. Failed tasks auto-retry with exponential backoff (1s → 2s → 4s → 8s → 16s).
4. After max retries, task escalates for manual review.

Distributed Tracing

- Every event gets a trace ID.
- As it flows through agents, each writes logs tagged with that ID.
- Search CloudWatch by trace ID to see the entire customer journey end-to-end.

State Management

Deal workflow:

1. `draft` → `submitted` → `kyc_pending` → `approved` → `funded` → `active`

Holiday logic:

1. `normal` → `holiday_pending` → `holiday_active` → `recovery_in_progress` → `normal`

Call Center (Twilio Flex)

When manual escalation is needed:

1. Ops team receives a ticket in Twilio Flex.
2. Agent clicks customer profile and Flex dials via Twilio.
3. Call recording logged automatically.
4. If customer calls in, IVR routes to available agent.
5. Agent sees customer profile, recent interactions, and account balance.

Sync vs. Async

Async by default (publish event, job queued). Sync only for external API calls:

- Stripe payment — 5-second timeout.
- IDology KYC — 10-second timeout.
- Property valuation — 15-second timeout.

All wrapped in circuit breaker pattern.

7. Interest Holiday Mechanism

Why holidays exist: Market downturns shouldn't force customers to pay lenders. We pause interest accrual automatically.

Threshold	What Happens
90% drop	Holiday automatically triggers. Customer still gets monthly distributions. Lender gets alert. Statements show holiday status.
145.8% recovery	Holiday automatically lifts. Everything returns to normal.

Example Timeline

- 1. **Months 1–10:** Normal operations. Fund performs well. Customer receives distributions.
- 2. **Month 11:** Market drop. Fund hits 88% (below 90% threshold). Holiday flag activates.
- 3. **Months 12–18:** Holiday active. Customer still paid. Statements note holiday status.
- 4. **Month 19:** Fund recovers to 145.8%. Holiday automatically released. Normal operations resume.

8. Compliance & Fraud Detection

Compliance Rules

Rule	What We Verify	What Happens If It Fails
KYC/AML	Customer identity confirmed before onboarding	Application rejected
Responsible Lending	Fund amount appropriate for risk tier	Margin increased; tier adjusted
Distribution Accuracy	Monthly calcs match fund returns	Distribution held pending audit
Fund Performance	Daily returns reconcile with actual	Discrepancy flagged
Holiday Logic	Thresholds applied correctly	Manually reviewed
Data Privacy	PII access logged; encrypted	Unauthorized access alert

Real-Time Fraud Monitoring

Account Anomalies

- Logins from unusual locations.
- Rapid device changes.
- After-hours access patterns.

Transaction Anomalies

- Multiple withdrawals in the same day.
- Requests immediately after distribution.
- Systematic holiday manipulation attempts.

Data Inconsistencies

- Reported vs. actual fund value mismatch.
- Distribution calculation errors.

Regulatory Violations

- KYC/AML failures.
- Undisclosed conflicts of interest.
- Unauthorized transfers.

Audit & Reporting

- **Audit Trail:** Every transaction, margin payment, and holiday trigger recorded with timestamp, user ID, and changes. Retained 7 years minimum.
- **Monthly Reporting:** Lenders and wholesale funders receive automated reports on portfolio health, margin collections, and default scenarios.
- **Annual Certification:** External auditor reviews all regulatory requirements (Australia, UK, US, NZ).

9. Multi-Jurisdiction Financial Regulation

FutureProof operates across four jurisdictions. Each has national financial regulation, state/territory-level requirements, and distinct licensing obligations. The platform enforces jurisdiction-specific rules automatically via the Compliance Agent and region configuration.

Australia

National Regulators & Legislation

Regulator / Act	Scope	FutureProof Obligation
ASIC	Consumer credit, financial services, market conduct	Australian Financial Services Licence (AFSL); Australian Credit Licence (ACL); responsible lending obligations
APRA	Prudential oversight of lenders and funders	Capital adequacy reporting for wholesale funders; liquidity coverage ratios
NCCP Act 2009	All consumer credit products	Responsible lending assessments; hardship obligations; mandatory disclosure; credit guides
AML/CTF Act 2006	Anti-money laundering, counter-terrorism financing	Customer identification (KYC); ongoing due diligence; suspicious matter reports (SMRs) to AUSTRAC; transaction monitoring
Corporations Act 2001	Financial product disclosure, managed investment schemes	Product Disclosure Statements (PDS) if EPM classified as financial product; ongoing disclosure obligations
Design & Distribution Obligations (DDO)	Product governance	Target Market Determination (TMD) for EPM product; distribution conditions; review triggers

State & Territory Requirements

State/Territory	Additional Requirement
NSW	Fair Trading Act 1987; Property and Stock Agents Act 2002
VIC	Australian Consumer Law and Fair Trading Act 2012; Estate Agents Act 1980
QLD	Fair Trading Act 1989; Property Occupations Act 2014
WA	Fair Trading Act 2010; Real Estate and Business Agents Act 1978

State/Territory	Additional Requirement
SA	Fair Trading Act 1987; Land Agents Act 1994
TAS	Australian Consumer Law (Tasmania) Act 2010
ACT	Fair Trading (Australian Consumer Law) Act 1992
NT	Consumer Affairs and Fair Trading Act 1990

United Kingdom

National Regulators & Legislation

Regulator / Act	Scope	FutureProof Obligation
FCA	Consumer protection, market integrity, competition	FCA authorisation; Consumer Duty compliance (2023); fair value assessments; vulnerability policies
PRA	Prudential soundness of financial firms	Capital requirements for lenders; stress testing; recovery and resolution plans
Consumer Credit Act 1974	Regulated credit agreements	Pre-contractual information; right of withdrawal (14 days); unfair relationship provisions
FSMA 2000	Overarching financial regulation	Permissions for regulated activities; approved persons regime; financial promotions
Money Laundering Regs 2017	AML/KYC	Customer due diligence; enhanced due diligence for high-risk; SARs to NCA
SM&CR	Individual accountability	Senior management functions defined; annual certification; conduct rules for all staff
Mortgage Credit Directive Order 2015	EU-derived mortgage regulation (retained)	Affordability assessments; ESIS; binding offers

Regional Considerations

Region	Additional Requirement
England & Wales	Law of Property Act 1925; Land Registration Act 2002
Scotland	Separate property law (feudal heritage); Registration of Title (Scotland) Act; separate Land Register
Northern Ireland	Property (NI) Order 1978; Land Registration Act (NI) 1970; separate registry

United States

Federal Regulators & Legislation

Regulator / Act	Scope	FutureProof Obligation
SEC	Securities regulation	Registration if EPM constitutes a security; Regulation D exemptions for private placements to wholesale funders
CFPB	Consumer financial products	TILA disclosures (APR, finance charges); RESPA settlement procedures; ability-to-repay rules
FinCEN	AML/BSA compliance	Currency Transaction Reports (CTRs); Suspicious Activity Reports (SARs); Customer Due Diligence (CDD) Rule
OCC	National bank oversight	Applicable if lender is nationally chartered; CRA compliance
TILA	Disclosure of credit terms	Loan Estimate within 3 business days; Closing Disclosure 3 days before settlement
RESPA	Settlement services	Good faith estimates; prohibition on kickbacks; escrow account requirements
ECOA	Anti-discrimination	Adverse action notices; prohibited basis protections; fair lending monitoring
Dodd-Frank Act	Systemic risk, consumer protection	Qualified Mortgage (QM) rules; ability-to-repay; risk retention for securitised products
BSA	AML compliance	AML programme requirement; independent testing; designated compliance officer

State-Level Requirements (Key States)

State	Licensing & Key Requirements
California	DFPI licence; California Financing Law (CFL); California Residential Mortgage Lending Act
New York	NYDFS mortgage banker licence; NY Banking Law Article 12-D; NYDFS cybersecurity regulation (23 NYCRR 500)
Texas	Department of Savings and Mortgage Lending licence; Texas Finance Code Chapter 156
Florida	Office of Financial Regulation mortgage lender licence; Florida Fair Lending Act
Illinois	IDFPR Residential Mortgage License; Illinois Residential Mortgage License Act
All 50 states	NMLS registration; state-specific usury caps; varying foreclosure procedures (judicial vs. non-judicial)

New Zealand

National Regulators & Legislation

Regulator / Act	Scope	FutureProof Obligation
FMA	Financial markets, services, conduct	Financial service provider registration; fair dealing obligations; licensed market conduct
RBNZ	Prudential oversight	Prudential requirements for non-bank deposit takers; macro-prudential policy (LVR restrictions)
CCCFA 2003	Consumer credit	Lender responsibility principles; affordability assessments; fee reasonableness; hardship provisions
CCCFA Amendments 2021	Strengthened responsible lending	Detailed income/expense verification; suitability assessments; borrower-centric affordability
FMCA 2013	Financial products, fair dealing	Fair dealing (misleading conduct); licensing for financial advice; disclosure obligations
AML/CFT Act 2009	Anti-money laundering	Customer due diligence; STRs to NZ Police FIU; annual AML/CFT risk assessment
FSP Act 2008	Provider registration	FSP registration; membership of approved dispute resolution scheme

Regional note: New Zealand operates as a unitary state — no state/territory-level financial regulation. All financial regulation is national. Local government consents may apply to property valuations under the Resource Management Act 1991.

10. IT Security & Privacy Regulation

Each jurisdiction imposes distinct data protection, privacy, and cybersecurity requirements. FutureProof maintains a unified security architecture that satisfies the strictest standard across all four markets.

Cross-Jurisdiction Security Standards

Standard	Scope	FutureProof Implementation
ISO 27001:2022	Information security management system	Certified ISMS covering all EPM data processing; annual surveillance audits; Statement of Applicability maintained
SOC 2 Type II	Trust service criteria (security, availability, confidentiality)	Annual audit by independent CPA firm; continuous control monitoring; report provided to wholesale funders
PCI DSS v4.0	Payment card data (if applicable)	Stripe handles PCI scope; SAQ-A compliance for redirect-based payments; no card data touches FutureProof servers
OWASP Top 10	Web application security	Brakeman (static analysis) in CI; annual penetration testing; WAF rules aligned to OWASP categories

Australia — Privacy & Cybersecurity

Regulation	Key Requirements	FutureProof Compliance
Privacy Act 1988 + APPs	Collection limitation; use/disclosure restrictions; data quality; security safeguards; cross-border disclosure rules	APP-compliant privacy policy; consent management; data minimisation; APP 8 cross-border transfer assessments
Notifiable Data Breaches (NDB)	Mandatory notification to OAIC and affected individuals	Incident response plan with 30-day assessment window; automated breach detection; notification templates pre-approved by legal
Consumer Data Right (CDR)	Open banking data sharing	CDR-compliant APIs if accessing banking data; accreditation as data recipient if required
APRA CPS 234	Information security for APRA-regulated entities	CPS 234 compliance evidence provided as data processor; security capability assessments; third-party assurance

Regulation	Key Requirements	FutureProof Compliance
Critical Infrastructure Act 2018 (SOCI)	Critical infrastructure protection	Risk management programme if designated; mandatory incident reporting within 12 hours for critical incidents

United Kingdom — Privacy & Cybersecurity

Regulation	Key Requirements	FutureProof Compliance
UK GDPR + DPA 2018	Lawful basis; data subject rights (access, erasure, portability); DPO requirement; DPIA; 72-hour breach notification to ICO	Lawful basis: contract + legitimate interest; automated DSAR portal; DPO appointed; DPIA completed; breach notification tested quarterly
FCA data requirements	SYSC 13 (operational risk); FG 16/5 (data security)	Operational resilience self-assessment; important business services identified; impact tolerances tested
NIS Regulations 2018	Network and information systems security	CAF (Cyber Assessment Framework) self-assessment; incident reporting to lead authority
UK Cyber Essentials Plus	Baseline cybersecurity certification	Certified annually; vulnerability scans; verified by external assessor
PSD2 / Open Banking	Strong Customer Authentication (SCA); secure API access	SCA for customer-facing transactions; OAuth 2.0 + FAPI-compliant API security

United States — Privacy & Cybersecurity

Regulation	Key Requirements	FutureProof Compliance
GLBA	Financial privacy; Safeguards Rule; pretexting protection	Privacy notices at onboarding and annually; written information security plan (WISP); vendor management programme
GLBA Safeguards Rule (2023)	Designated qualified individual; risk assessment; encryption; MFA; continuous monitoring	Qualified individual designated; annual risk assessment; AES-256 at rest + TLS 1.3 in transit; MFA enforced; SIEM-based monitoring
NYDFS 23 NYCRR 500	Comprehensive cybersecurity for NY financial services	Cybersecurity programme; CISO appointed; 72-hour incident notification; annual pen testing; 500.17 certification filed annually
CCPA / CPRA (California)	Consumer privacy rights; right to delete; right to opt-out; data minimisation	California-specific privacy notices; opt-out mechanisms; data inventory maintained; annual CPRA risk assessments
State privacy laws	Virginia (VCDPA), Colorado (CPA), Connecticut (CTDPA), Utah (UCPA), Texas (TDPSA), Oregon, Montana, etc.	Unified privacy framework satisfying strictest state standard; configurable consent flows per state; vendor DPA library
SEC Cybersecurity Rules (2023)	Material incident disclosure within 4 business days (8-K); annual risk disclosure (10-K)	Incident materiality assessment process; board-level cybersecurity governance documented

State Breach Notification

All 50 US states have breach notification laws with varying requirements:

Requirement	Range Across States
Notification timeline	30 days (Florida, Colorado) to 90 days (Connecticut) to "most expedient time" (majority)
AG notification	Required in ~35 states above certain thresholds (typically 500+ individuals)
Credit monitoring	Required in some states (e.g., California for SSN; Massachusetts)
Definition of personal information	Varies — some include biometrics, geolocation, health data, online credentials

FutureProof implements the strictest standard: 30-day notification, AG notification for all breaches >250 individuals, free credit monitoring offered for any breach involving financial data.

New Zealand — Privacy & Cybersecurity

Regulation	Key Requirements	FutureProof Compliance
Privacy Act 2020	13 Information Privacy Principles (IPPs); mandatory breach notification; cross-border disclosure rules	IPP-compliant data handling; Privacy Officer appointed; breach notification to OPC within 72 hours; cross-border transfer assessments
Notifiable Privacy Breaches	Mandatory notification to OPC and affected individuals for breaches causing serious harm	Automated breach assessment workflow; pre-approved notification templates; harm assessment matrix
RBNZ Cybersecurity Guidelines	Cyber resilience expectations for financial institutions	Annual cyber resilience self-assessment; incident response tested; third-party risk management
CERT NZ reporting	Voluntary but expected incident reporting	Integrated into incident response process; CERT NZ contacted for significant incidents
NZISM	Government security standard	Aligned to NZISM controls where applicable; used as security baseline

Data Residency & Cross-Border Transfers

Jurisdiction	Data Residency Rule	FutureProof Approach
Australia	APP 8 requires reasonable steps before cross-border disclosure; APRA CPS 234 expects control	Primary data in AU region (ap-southeast-2); cross-border transfers only with APP 8 assessment and contractual safeguards
UK	UK GDPR adequacy decisions or appropriate safeguards (UK IDTA / UK SCCs)	Primary data in UK region (eu-west-2); transfers to adequate countries only; UK IDTA for others
US	No federal data residency law; GLBA requires safeguards regardless of location	Primary data in US region (us-east-1); state-specific restrictions enforced; GLBA safeguards applied globally
NZ	Privacy Act 2020 IPP 12 — disclosure to foreign entity only if comparable protections exist	Primary data in AU region (ap-southeast-2) with NZ-specific encryption keys; cross-border assessment completed

Platform Security Architecture

The following security controls apply across all jurisdictions:

- Encryption at rest:** AES-256 for all databases, S3 buckets, and backups.
- Encryption in transit:** TLS 1.3 minimum for all connections; certificate pinning for inter-service communication.
- Authentication:** MFA enforced for all internal access; OAuth 2.0 + PKCE for customer-facing; hardware security keys for admin.
- Authorisation:** RBAC with principle of least privilege; ABAC for cross-jurisdiction data access.
- Key management:** AWS KMS with per-jurisdiction CMKs; automatic key rotation every 365 days; HSM-backed for signing.
- Logging & audit:** All access logged with user ID, IP, timestamp, action; logs immutable (append-only); retained per jurisdiction requirements (7 years AU/UK/NZ, varies by state in US).
- Penetration testing:** Annual external pen test; quarterly vulnerability scans; bug bounty programme.
- Incident response:** Documented IR plan; tested quarterly; jurisdiction-specific notification timelines enforced.
- Vendor management:** All third-party vendors assessed against SOC 2 or ISO 27001; DPAs in place; annual reassessment.
- Business continuity:** RPO < 1 hour; RTO < 4 hours; multi-AZ deployment; tested failover twice annually.

11. Technology Stack

Layer	Technology
Backend	Rails 8.1 (Ruby 3.4.8), Solid Queue (4 workers), ruby-kafka, Puma (3 threads, auto-scaling)
Data	PostgreSQL 16 (primary + read replica), Redis 7, Kafka 3.6 (3 replicas), PgBouncer
Agents & Workflow	Solid Queue, Kafka pub/sub (4 topics), OpenTelemetry, dead-letter queues, state machines
Communication	SendGrid (email), Twilio (SMS + voice + Flex), WebRTC, DocuSign
Frontend	Hotwire (Turbo Streams + Drive), Stimulus JS, Tailwind CSS, ERB, Chart.js
Infrastructure	AWS S3/CloudFront/Glacier, Fly.io, GitHub Actions, Docker, Terraform
Monitoring	Sentry, CloudWatch, Grafana, PagerDuty, HTTPS/TLS, WAF, CSP headers

12. Database Schema

Core Tables

Table	Purpose	Key Fields
customers	Homeowners	id, email, phone, kyc_status, aml_status, risk_tier
properties	Residential with equity	id, customer_id, address, property_value, equity_percentage, loan_amount
accounts	Customer investment accounts	id, customer_id, property_id, starting_balance, current_balance, holiday_flagged
fund_performance	Monthly metrics (materialized view)	account_id, month, starting_value, ending_value, return_pct, rolling_12m_return
distributions	Monthly payouts (immutable)	id, account_id, fund_return_amount, lender_margin, customer_payout, executed_at
payments	ACH/bank transfers (immutable)	id, distribution_id, amount, bank_account_id, status, stripe_charge_id
audit_log	Append-only event log	id, entity_type, action, changes_json, user_id, timestamp, trace_id

Table	Purpose	Key Fields
fraud_alerts	Detected patterns	id, customer_id, alert_type, fraud_score, status, reviewed_by
lenders	Loan originators	id, name, tier, max_funding_line, current_balance, margin_rate
wholesale_funders	Capital providers	id, name, funding_capacity, current_deployed, margin_rate

Design Patterns

- **Immutable logs:** distributions , payments , audit_log have no UPDATE/DELETE. All changes are INSERT-only with timestamps.
- **Soft deletes:** customers , accounts include deleted_at . Preserves history; queries filter WHERE deleted_at IS NULL .
- **JSON columns:** audit_log.changes_json stores before/after. fraud_alerts.details_json stores model output. accounts.metadata_json stores region-specific terms.
- **Materialized views:** fund_performance refreshed nightly. Improves dashboard load times.
- **Partitioning:** distributions , payments partitioned by created_at (monthly partitions). Old partitions move to cold storage.

13. Deployment & Quality Assurance

Testing

- **Unit tests:** Core logic (distributions, holiday logic, fraud rules).
- **Integration tests:** Agent workflows, end-to-end customer journeys, payments.
- **Performance tests:** Batch processing at scale (1000s of customers), query optimization.
- **Security tests:** Auth, authorization, data access, PII encryption.
- **Manual tests:** Customer portal UX, lender reporting, admin workflows.

Continuous Deployment

Pull Requests

1. GitHub Actions runs tests on every PR.
2. RSpec (200+ specs), Rubocop (linting), Brakeman (security) all pass.
3. Database migrations validated.
4. Code review approval required before merge.

Staging

1. Deployed automatically from main branch using Fly.io (same infra as production).
2. Runs against production-replica DB (encrypted, daily sanitised PII).
3. Manual smoke tests before promotion.

Production

1. Blue-green deployment on Fly.io.
2. Traffic shifted: 10% → 50% → 100% over 10 minutes (canary rollout).
3. Health checks verify Rails boot, database, Kafka, and latency.
4. Automatic rollback if health checks fail within 5 minutes.

Monitoring & Alerting

- **Sentry:** Real-time errors. Critical errors trigger PagerDuty within 2 minutes.
- **CloudWatch:** Structured JSON logging (trace_id, user_id, duration_ms, queries, API calls). 90 days hot, then S3 archive.

- **Grafana:** System health, app metrics, business metrics, agent health, database health.
 - **Alert thresholds:** Error rate >1%, distribution latency >60s, Kafka lag >1000 messages.
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